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Addis Ababa University
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**COLLEGE OF NATURAL AND COMPUTATIONAL
SCIENCES**

DEPARTMENT OF COMPUTER SCIENCE

Applied Numerical Analysis Assignment

Thomas Method to solve tridiagonal system of Equations

Prepared by:

<u>ID</u>	<u>Name</u>
UGR/5588/16	Eyob Nebyou
UGR/9805/15	Gizachew Mohammed
UGR/7248/16	Henok Tadesse
UGR/0880/16	Getaneh Dachew
UGR/4913/16	Leawi Taddesse
UGR/1219/16	Eyob Assayie
UGR/0796/15	Adonias Abiyot
UGR/8107/17	Lawgaw sefineh

Instructor: Dr Temesgen A.
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Thomas method (to solve tridiagonal system of equations)

```
function x = thomas_method(a, b, c, d)
% Thomas method for tridiagonal system
% a : sub-diagonal (a(1) unused)
% b : main diagonal
% c : super-diagonal (c(n) unused)
% d : right-hand side

n = length(b);

% Forward elimination
for i = 2:n
    m = a(i) / b(i-1);
    b(i) = b(i) - m * c(i-1);
    d(i) = d(i) - m * d(i-1);
end

% Back substitution
x = zeros(n,1);
x(n) = d(n) / b(n);

for i = n-1:-1:1
    x(i) = (d(i) - c(i)*x(i+1)) / b(i);
end
end
```